

A guide to the arithmetic certificate for $K = \mathbf{Q}(\sqrt{-75949255})$, $p = 5$

Generated by `tools/certificate_guide.gp`

What this document is

This guide restates the content of `certificate.gp` in the notation of Section 2.5 of the paper. Each of the 18 entries certifies one value $D_x(e_j)$ of a secondary norm operator: it stores the unramified cyclic quintic L_x/K , the normalized generator σ_x , the pair (a', J) representing the torsion class e_j , the auxiliary divisor I' , the compactly represented element t , a residue prime for the sign check, and the expected norm class $[N_x(I')]$. The two defining equations are

$$(1 - \sigma_x)^2 I' + \text{div}(t) + i_x(J) = 0, \quad N_x(t) = a'.$$

Facts marked **Checked** are recomputed exactly by the generating script; the full verification is performed by the shared verifier `certificate/verify_certificate.c` as described in the README files of the `certificate` directory.

Notation and conventions

Ideals as matrices. An ideal of the ring of integers is stored by its *Hermite normal form* (HNF): an upper-triangular integer matrix whose columns are the coordinates, with respect to the fixed integral basis, of a $\mathbf{Z}$ -basis of the ideal. For example, the ideal $J = \begin{pmatrix} 256 & 21 \\ 0 & 1 \end{pmatrix}$ of Entry 1 below, over K with integral basis $(1, s)$, denotes the ideal $\mathbf{Z}256 + \mathbf{Z}(21 + s)$. The determinant of the matrix is the norm of the ideal. Elements are likewise given by their coordinate vectors in the integral basis.

Characters. The header fixes three generators $\kappa_1, \kappa_2, \kappa_3$ of $\text{Cl}(K)$ (as HNF ideals). A character x on $\text{Cl}(K)/5$ is recorded by its value vector $(x(\bar{\kappa}_1), x(\bar{\kappa}_2), x(\bar{\kappa}_3))$; thus $(1, 0, 0)$ is the character dual to $\bar{\kappa}_1$. The column number j records which basis element e_j of $\text{Cl}(K)[5]$ the stored pair (a', J) represents.

Automorphisms. σ_x is a field automorphism of L_x . Writing θ for the chosen root of the stored absolute polynomial, so that $L_x = \mathbf{Q}(\theta)$, the automorphism is determined by the single value $\sigma_x(\theta)$, and the certificate stores the coordinate vector of $\sigma_x(\theta)$ in the integral basis of L_x .

Base field data

The base field is $K = \mathbf{Q}(s)$ with $s^2 - s + 18987314 = 0$, of discriminant -75949255 . Class group invariants $(40 \ 20 \ 5)$, class number 4000; the three stored generator ideals (HNF) are $\begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}$, $\begin{pmatrix} 22 & 13 \\ 0 & 1 \end{pmatrix}$, $\begin{pmatrix} 4114 & 607 \\ 0 & 1 \end{pmatrix}$, and the torsion units are generated by -1 of order 2.

Entry 1: character a , column e_1

Prescribed character vector $(1 \ 0 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 + (-s + 1426)y^3 + (27s - 107905)y^2 + (-238s + 1392475)y + (1088s - 4891799)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 2y^9 + 2852y^8 - 218634y^7 + 24019859y^6 - 1345480831y^5 \\ & + 38499678307y^4 - 599733073788y^3 + 5185164965896y^2 - 23454038 \\ & 436920y + 46400494202705 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 0 \ -1 \ 0 \ 0 \ 1 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{477889}{549755813888} + \left(-\frac{99}{1099511627776}\right)s, \quad J = \begin{pmatrix} 256 & 21 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 256$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $3230601672625 = 5^3 \cdot 25844813381$:

$$\begin{pmatrix} 129224066905 & 69690603515 & 45718187850 & 106863314831 & 20146063844 & 83273874827 & 106163847728 & 114789602833 & 112945233567 & 13070778232 \\ 0 & 5 & 0 & 0 & 1 & 0 & 2 & 0 & 0 & 3 \\ 0 & 0 & 5 & 4 & 2 & 3 & 0 & 2 & 0 & 3 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 502 factors with signed exponents of absolute value up to 1756313213800515074953828174; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (0 \ 3 \ 2)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 2: character a , column e_2

Prescribed character vector $(1 \ 0 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 + (-s + 1426)y^3 + (27s - 107905)y^2 + (-238s + 1392475)y + (1088s - 4891799)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 2y^9 + 2852y^8 - 218634y^7 + 24019859y^6 - 1345480831y^5 \\ & + 38499678307y^4 - 599733073788y^3 + 5185164965896y^2 - 23454038 \\ & 436920y + 46400494202705 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 0 \ -1 \ 0 \ 0 \ 1 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{385737}{7878081286000} + \left(\frac{43}{15756162572000}\right)s, \quad J = \begin{pmatrix} 830 & 47 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 830$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $1570108656483 = 3^2 \cdot 174456517387$:

$$\begin{pmatrix} 523369552161 & 517859494083 & 453125366563 & 499108715932 & 324933075922 & 508326874712 & 324935153520 & 218404431650 & 282732652252 & 13041597718 \\ 0 & 3 & 2 & 0 & 1 & 0 & 2 & 0 & 0 & 2 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 503 factors with signed exponents of absolute value up to 1527548714546768827385452862; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (0 \ 1 \ 3)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 3: character a , column e_3

Prescribed character vector $(1 \ 0 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 + (-s + 1426)y^3 + (27s - 107905)y^2 + (-238s + 1392475)y + (1088s - 4891799)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 2y^9 + 2852y^8 - 218634y^7 + 24019859y^6 - 1345480831y^5 \\ + 38499678307y^4 - 599733073788y^3 + 5185164965896y^2 - 23454038 \\ 436920y + 46400494202705$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 0 \ -1 \ 0 \ 0 \ 1 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{460074979}{589238942107232912} + \left(\frac{132217}{1178477884214465824}\right)s, \quad J = \begin{pmatrix} 4114 & 607 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 4114$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $5527782070875 = 3^2 \cdot 5^3 \cdot 4913584063$:

$$\begin{pmatrix} 73703760945 & 19872026850 & 20064345955 & 3053963745 & 53615380030 & 40059094027 & 2925080248 & 56887145479 & 33553696778 & 39956718757 \\ 0 & 15 & 10 & 11 & 9 & 10 & 0 & 3 & 10 & 11 \\ 0 & 0 & 5 & 2 & 3 & 3 & 3 & 3 & 3 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 503 factors with signed exponents of absolute value up to 1221548812224684959661120760; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of +1. Certified value: $D_x(e_3) = (0 \ 0 \ 1)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 4: character b , column e_1

Prescribed character vector $(0 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 + (-s + 1386)y^3 + (-19s + 160910)y^2 + (64s + 2446992)y + (1476s + 9253032)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 2y^9 + 2772y^8 + 319030y^7 + 25479171y^6 + 1180986700y^5 \\ + 37075210968y^4 + 710869269664y^3 + 7978610479856y^2 + 48875629 \\ 773120y + 126997565253120$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \ 0 \ 0 \ 0 \ 0 \ -1 \ 0 \ 1 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{477889}{549755813888} + \left(-\frac{99}{1099511627776}\right) s, \quad J = \begin{pmatrix} 256 & 21 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 256$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $59110480752640 = 2^{12} \cdot 5 \cdot 127 \cdot 22726409$:

$$\begin{pmatrix} 115450157720 & 58604017300 & 58248579064 & 96476249458 & 14822652364 & 73070689970 & 63347750813 & 44055279435 & 105709304015 & 106048415133 \\ 0 & 4 & 0 & 2 & 2 & 2 & 3 & 2 & 2 & 0 \\ 0 & 0 & 4 & 0 & 2 & 0 & 1 & 2 & 0 & 2 \\ 0 & 0 & 0 & 4 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 2 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 186 factors with signed exponents of absolute value up to 29464150824522635; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (2 \ 0 \ 4)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 5: character b , column e_2

Prescribed character vector $(0 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 + (-s + 1386)y^3 + (-19s + 160910)y^2 + (64s + 2446992)y + (1476s + 9253032)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 2y^9 + 2772y^8 + 319030y^7 + 25479171y^6 + 1180986700y^5 \\ & + 37075210968y^4 + 710869269664y^3 + 7978610479856y^2 + 48875629 \\ & 773120y + 126997565253120 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \ 0 \ 0 \ 0 \ 0 \ -1 \ 0 \ 1 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{385737}{7878081286000} + \left(\frac{43}{15756162572000}\right) s, \quad J = \begin{pmatrix} 830 & 47 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 830$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $32430087103328 = 2^5 \cdot 29 \cdot 34946214551$:

$$\begin{pmatrix} 4053760887916 & 1114975850816 & 3918827920193 & 2377368419319 & 2189792053454 & 3996667697014 & 985170487046 & 2282676270954 & 369700578806 & 3688444562001 \\ 0 & 2 & 1 & 1 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 190 factors with signed exponents of absolute value up to 32988939360536606; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (3 \ 0 \ 4)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 6: character b , column e_3

Prescribed character vector $(0 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 + (-s + 1386)y^3 + (-19s + 160910)y^2 + (64s + 2446992)y + (1476s + 9253032)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 2y^9 + 2772y^8 + 319030y^7 + 25479171y^6 + 1180986700y^5 \\ & + 37075210968y^4 + 710869269664y^3 + 7978610479856y^2 + 48875629 \\ & 773120y + 126997565253120 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \ 0 \ 0 \ 0 \ 0 \ -1 \ 0 \ 1 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{460074979}{589238942107232912} + \left(\frac{132217}{1178477884214465824}\right)s, \quad J = \begin{pmatrix} 4114 & 607 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 4114$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $28813429757440 = 2^9 \cdot 5 \cdot 29 \cdot 103 \cdot 3768077$:

$$\begin{pmatrix} 450209839960 & 198012576000 & 322992154428 & 60039223482 & 158287608112 & 427185093359 & 117977687083 & 292868893808 & 315076651153 & 138173422662 \\ 0 & 4 & 0 & 2 & 3 & 1 & 3 & 0 & 1 & 3 \\ 0 & 0 & 4 & 0 & 1 & 1 & 0 & 1 & 3 & 1 \\ 0 & 0 & 0 & 2 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 185 factors with signed exponents of absolute value up to 22981542407621309; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (2 \ 0 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 7: character c , column e_1

Prescribed character vector $(0 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 4y^4 - 1019y^3 - 37334y^2 + (-288s - 2488676)y + (16320s - 33863400)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 8y^9 - 2022y^8 - 66516y^7 - 3640607y^6 + 28286772y^5 + 6 \\ & 736884636y^4 + 254832190880y^3 + 10297011804400y^2 - 996762835680 \\ & 0y + 6203303989185600 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \ -1 \ 0 \ 1 \ -1 \ 1 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{477889}{549755813888} + \left(-\frac{99}{1099511627776}\right)s, \quad J = \begin{pmatrix} 256 & 21 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 256$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $389865520000 = 2^7 \cdot 5^4 \cdot 11 \cdot 43 \cdot 10303$:

$$\begin{pmatrix} 97466380 & 7464400 & 15051180 & 74877597 & 83897096 & 71113476 & 29447433 & 71411485 & 93973096 & 50071820 \\ 0 & 20 & 0 & 3 & 2 & 18 & 1 & 5 & 1 & 7 \\ 0 & 0 & 10 & 4 & 4 & 8 & 7 & 0 & 7 & 4 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 5 & 3 & 4 & 4 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 126 factors with signed exponents of absolute value up to 3662; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 17$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (4 \ 2 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 8: character c , column e_2

Prescribed character vector $(0 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 4y^4 - 1019y^3 - 37334y^2 + (-288s - 2488676)y + (16320s - 33863400)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 8y^9 - 2022y^8 - 66516y^7 - 3640607y^6 + 28286772y^5 + 6 \\ & 736884636y^4 + 254832190880y^3 + 10297011804400y^2 - 996762835680 \\ & 0y + 6203303989185600 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \ -1 \ 0 \ 1 \ -1 \ 1 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{385737}{7878081286000} + \left(\frac{43}{15756162572000}\right)s, \quad J = \begin{pmatrix} 830 & 47 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 830$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $6956054458560 = 2^6 \cdot 3^2 \cdot 5 \cdot 2415296687$:

$$\begin{pmatrix} 289835602440 & 172671645387 & 101843854436 & 106943730740 & 92951153616 & 42442331118 & 74702172028 & 224614252324 & 31249907107 & 28319714887 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 8 & 5 & 5 & 5 & 3 & 0 \\ 0 & 0 & 0 & 0 & 0 & 3 & 2 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 157 factors with signed exponents of absolute value up to 1377007445; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 17$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of +1. Certified value: $D_x(e_2) = (1 \ 0 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 9: character c , column e_3

Prescribed character vector $(0 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 4y^4 - 1019y^3 - 37334y^2 + (-288s - 2488676)y + (16320s - 33863400)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 8y^9 - 2022y^8 - 66516y^7 - 3640607y^6 + 28286772y^5 + 6 \\ & 736884636y^4 + 254832190880y^3 + 10297011804400y^2 - 996762835680 \\ & 0y + 6203303989185600 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \ -1 \ 0 \ 1 \ -1 \ 1 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{460074979}{589238942107232912} + \left(\frac{132217}{1178477884214465824}\right)s, \quad J = \begin{pmatrix} 4114 & 607 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 4114$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $41878023644160 = 2^{10} \cdot 3^2 \cdot 5 \cdot 11^2 \cdot 1487 \cdot 5051$:

$$\begin{pmatrix} 109057353240 & 46989926744 & 92503465724 & 7378677500 & 99132730864 & 45841353691 & 35433155714 & 8179390664 & 11226486000 & 31630729491 \\ 0 & 2 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 4 & 0 & 0 & 0 & 2 & 0 & 1 & 0 \\ 0 & 0 & 0 & 4 & 0 & 0 & 0 & 3 & 3 & 2 \\ 0 & 0 & 0 & 0 & 2 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 3 & 0 & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 126 factors with signed exponents of absolute value up to 14954119; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 17$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of +1. Certified value: $D_x(e_3) = (2 \ 2 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 10: character $a + b$, column e_1

Prescribed character vector $(1 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 + (-s + 1391)y^3 + (-10s - 53867)y^2 + (17s + 497178)y + (-1716s + 4919343)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 2y^9 + 2782y^8 - 110525y^7 + 22022921y^6 + 238770840y^5 \\ & + 5528193778y^4 + 18818218644y^3 + 374389109516y^2 + 378301285147 \\ & 5y + 80102602053045 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ -1 \ 0 \ 0 \ 0 \ -1 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{477889}{549755813888} + \left(-\frac{99}{1099511627776}\right)s, \quad J = \begin{pmatrix} 256 & 21 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 256$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $582174874454229 = 3^2 \cdot 7^2 \cdot 43 \cdot 61^2 \cdot 139 \cdot 59357$:

$$\begin{pmatrix} 454469066709 & 418325050560 & 144968472069 & 104469683806 & 132469198262 & 221925824869 & 148946619689 & 14994517432 & 166511600926 & 131385562853 \\ 0 & 3 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 \\ 0 & 0 & 61 & 18 & 0 & 0 & 6 & 28 & 12 & 30 \\ 0 & 0 & 0 & 7 & 0 & 0 & 1 & 2 & 6 & 5 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 338 factors with signed exponents of absolute value up to 8812814898; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 17$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (4 \ 1 \ 1)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 11: character $a + b$, column e_2

Prescribed character vector $(1 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 + (-s + 1391)y^3 + (-10s - 53867)y^2 + (17s + 497178)y + (-1716s + 4919343)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 2y^9 + 2782y^8 - 110525y^7 + 22022921y^6 + 238770840y^5 \\ & + 5528193778y^4 + 18818218644y^3 + 374389109516y^2 + 378301285147 \\ & 5y + 80102602053045 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ -1 \ 0 \ 0 \ 0 \ -1 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{385737}{7878081286000} + \left(\frac{43}{15756162572000}\right)s, \quad J = \begin{pmatrix} 830 & 47 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 830$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $700240148927 = 11 \cdot 63658195357$:

$$\begin{pmatrix} 700240148927 & 338912989232 & 274565176170 & 286232192234 & 235259970704 & 435050337186 & 590485335340 & 318574028349 & 23207445936 & 597468153979 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 336 factors with signed exponents of absolute value up to 7309133151; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 17$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (2 \ 3 \ 4)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 12: character $a + b$, column e_3

Prescribed character vector $(1 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 + (-s + 1391)y^3 + (-10s - 53867)y^2 + (17s + 497178)y + (-1716s + 4919343)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 2y^9 + 2782y^8 - 110525y^7 + 22022921y^6 + 238770840y^5 \\ + 5528193778y^4 + 18818218644y^3 + 374389109516y^2 + 378301285147 \\ 5y + 80102602053045$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ -1 \ 0 \ 0 \ 0 \ -1 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{460074979}{589238942107232912} + \left(\frac{132217}{1178477884214465824}\right)s, \quad J = \begin{pmatrix} 4114 & 607 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 4114$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $12400947627875 = 5^3 \cdot 137941 \cdot 719203$:

$$\begin{pmatrix} 496037905115 & 2226323635 & 227534672730 & 342021681910 & 157748075547 & 422233141292 & 222923645571 & 388960118755 & 435359630678 & 478127784280 \\ 0 & 5 & 0 & 4 & 4 & 1 & 2 & 2 & 0 & 2 \\ 0 & 0 & 5 & 4 & 0 & 0 & 2 & 2 & 3 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 350 factors with signed exponents of absolute value up to 10369723651; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 17$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (2 \ 3 \ 1)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 13: character $a + c$, column e_1

Prescribed character vector $(1 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 3180y^3 + (26s - 13)y^2 + 587504y + (-1438s + 719)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 6360y^8 + 11287408y^6 + 9098898655y^4 - 1074634422954y^2 \\ + 39262802814055$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 1 \ -1 \ -1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{477889}{549755813888} + \left(-\frac{99}{1099511627776}\right) s, \quad J = \begin{pmatrix} 256 & 21 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 256$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $16933696425491 = 19^2 \cdot 2393 \cdot 19602067$:

$$\begin{pmatrix} 891247180289 & 317727667181 & 404559778389 & 368993959401 & 176816454819 & 768037218640 & 230573207231 & 556680766635 & 392552709757 & 775171687987 \\ 0 & 19 & 0 & 0 & 12 & 0 & 6 & 6 & 18 & 14 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 655 factors with signed exponents of absolute value up to 37468427207710989; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (2 \ 4 \ 3)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 14: character $a + c$, column e_2

Prescribed character vector $(1 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 3180y^3 + (26s - 13)y^2 + 587504y + (-1438s + 719)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 6360y^8 + 11287408y^6 + 9098898655y^4 - 1074634422954y^2 + 39262802814055$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 1 \ -1 \ -1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{385737}{7878081286000} + \left(\frac{43}{15756162572000}\right) s, \quad J = \begin{pmatrix} 830 & 47 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 830$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $237822829488235 = 5 \cdot 389 \cdot 1733 \cdot 70556231$:

$$\begin{pmatrix} 237822829488235 & 71205069741772 & 98654025112227 & 229416477467993 & 149955228543615 & 127768159339434 & 30462909934714 & 13827174926858 & 147450532751207 & 127671698738658 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 655 factors with signed exponents of absolute value up to 8005810598658165598; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (3 \ 0 \ 2)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 15: character $a + c$, column e_3

Prescribed character vector $(1 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 3180y^3 + (26s - 13)y^2 + 587504y + (-1438s + 719)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 6360y^8 + 11287408y^6 + 9098898655y^4 - 1074634422954y^2 + 39262802814055$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 1 \ -1 \ -1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{460074979}{589238942107232912} + \left(\frac{132217}{1178477884214465824}\right)s, \quad J = \begin{pmatrix} 4114 & 607 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 4114$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $634801694441 = 103 \cdot 6163123247$:

$$\begin{pmatrix} 634801694441 & 516838584540 & 531245804914 & 186833454937 & 321737297434 & 51682868569 & 394772994018 & 219876233523 & 150488485830 & 245194352089 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 656 factors with signed exponents of absolute value up to 4658440687261638698; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (1 \ 2 \ 4)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 16: character $b + c$, column e_1

Prescribed character vector $(0 \ 1 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 4897y^3 + (-4s + 2)y^2 + 11871356y + (54096s - 27048)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} + 9794y^8 + 47723321y^6 + 116571857684y^4 + 132711991481776y^2 + 55564042351043520$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(2 \ -1 \ 1 \ -1 \ 2 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{477889}{549755813888} + \left(-\frac{99}{1099511627776}\right)s, \quad J = \begin{pmatrix} 256 & 21 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 256$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $17084375562240 = 2^{11} \cdot 3^2 \cdot 5 \cdot 7^2 \cdot 3783211$:

$$\begin{pmatrix} 3177897240 & 1856506876 & 1867829572 & 1040419598 & 2325445912 & 1078259096 & 2907319822 & 2091901363 & 1645593617 & 2481696554 \\ 0 & 28 & 16 & 14 & 0 & 10 & 0 & 26 & 5 & 11 \\ 0 & 0 & 12 & 0 & 0 & 4 & 0 & 2 & 2 & 7 \\ 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 212 factors with signed exponents of absolute value up to 28087026; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (2 \ 3 \ 2)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 17: character $b + c$, column e_2

Prescribed character vector $(0 \ 1 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 4897y^3 + (-4s + 2)y^2 + 11871356y + (54096s - 27048)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} + 9794y^8 + 47723321y^6 + 116571857684y^4 + 132711991481776y^2 + 55564042351043520$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(2 \ -1 \ 1 \ -1 \ 2 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{385737}{7878081286000} + \left(\frac{43}{15756162572000}\right)s, \quad J = \begin{pmatrix} 830 & 47 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 830$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $52321878016000 = 2^{18} \cdot 5^3 \cdot 1596737$:

$$\begin{pmatrix} 8175293440 & 1965789460 & 7985385540 & 1259530504 & 7786920346 & 2036639997 & 7385355151 & 6709827276 & 582576653 & 6619406215 \\ 0 & 20 & 0 & 2 & 12 & 15 & 19 & 14 & 16 & 2 \\ 0 & 0 & 20 & 6 & 6 & 8 & 2 & 4 & 18 & 9 \\ 0 & 0 & 0 & 2 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 293 factors with signed exponents of absolute value up to 684541021; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (3 \ 1 \ 4)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 18: character $b + c$, column e_3

Prescribed character vector $(0 \ 1 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 4897y^3 + (-4s + 2)y^2 + 11871356y + (54096s - 27048)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} + 9794y^8 + 47723321y^6 + 116571857684y^4 + 132711991481776y^2 + 55564042351043520$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(2 \ -1 \ 1 \ -1 \ 2 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{460074979}{589238942107232912} + \left(\frac{132217}{1178477884214465824}\right)s, \quad J = \begin{pmatrix} 4114 & 607 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 4114$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $231874899712000 = 2^{11} \cdot 5^3 \cdot 905761327$:

$$\begin{pmatrix} 36230453080 & 2972203880 & 13035000400 & 36111800288 & 3924739390 & 4538883497 & 14745563253 & 4977424729 & 15568102553 & 18817149908 \\ 0 & 10 & 0 & 4 & 4 & 5 & 2 & 3 & 6 & 1 \\ 0 & 0 & 20 & 12 & 12 & 8 & 15 & 18 & 18 & 3 \\ 0 & 0 & 0 & 4 & 2 & 1 & 2 & 3 & 0 & 3 \\ 0 & 0 & 0 & 0 & 2 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 212 factors with signed exponents of absolute value up to 88791; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (4 \ 2 \ 3)$ in the coordinates of $\text{Cl}(K)/5$.